MINI PROJECT REPORT

ON

AI FARMER DECISION ASSISTANT

An AI-Assisted Web Application for Agricultural Decision Support

Submitted in partial fulfillment of the requirements for the degree of

B.Tech – Computer Science and Engineering (Artificial Intelligence & Machine Learning)

Submitted By

Name: _____

Roll No: _____

III B.Tech – CSE (AI & ML)

Under the Guidance of

Guide Name: _____

Department of CSE (AI & ML)

College Name: _____

Academic Year: 2026–2027

CERTIFICATE

This is to certify that the mini project entitled “AI Farmer Decision Assistant” is a bonafide work carried out by _____, a student of III B.Tech – CSE (AI & ML), during the academic year 2026–2027, under the guidance and supervision of the undersigned. The project has been developed as part of the academic requirements for the mini project.

Project Guide Signature

Head of the Department

Date: _____ Place: _____

DECLARATION

I hereby declare that the mini project entitled “AI Farmer Decision Assistant” is my original work carried out as part of my B.Tech CSE (AI & ML) curriculum. The project uses the Base44 AI-assisted development platform to create a web-based agricultural decision-support application. The work has been completed under the guidance of the project supervisor and has not been submitted elsewhere for any other academic award.

Signature of Student

Name: _____

Date: _____

ACKNOWLEDGEMENT

I express my sincere gratitude to my project guide, faculty members, and the Department of Computer Science and Engineering (AI & ML) for their valuable guidance and support throughout the development of this mini project. I also thank my institution for providing the facilities and learning environment needed to complete the project. Finally, I thank my friends and family for their encouragement and support.

ABSTRACT

Agriculture plays an important role in the Indian economy, but farmers often face difficulties while making decisions related to crop selection, irrigation, fertilizers, weather conditions, crop problems, and farming practices. The AI Farmer Decision Assistant is an AI-assisted web application designed to provide simple and understandable agricultural decision support. Farmers can enter information such as crop, soil type, location, water availability, weather condition, and crop growth stage. The application processes these inputs and generates useful suggestions related to crop selection, irrigation, fertilizer management, crop care, and possible crop problems. The project is developed using the Base44 AI tool, which supports rapid AI-assisted web application development. The system is designed with a simple, responsive, and farmer-friendly interface. The project demonstrates how artificial intelligence and modern web technologies can be applied to support agricultural decision-making.

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1. INTRODUCTION

Agriculture is one of the most important sectors in India. Farmers need to make several decisions throughout the farming process, including selecting suitable crops, determining irrigation requirements, choosing fertilizers, identifying crop problems, and understanding weather conditions. Traditional decision-making often depends on personal experience and local knowledge. While these methods are valuable, farmers may not always have timely access to personalized information. Artificial Intelligence can support this process by analyzing agricultural inputs and providing understandable recommendations. The AI Farmer Decision Assistant provides a simple platform where farmers can enter basic information and receive AI-assisted suggestions.

2. PROBLEM STATEMENT

Farmers frequently face difficulties in making timely and informed agricultural decisions because of limited access to personalized agricultural information, uncertainty about crop selection and irrigation, difficulty identifying common crop problems, and dependence on general advice. There is a need for a simple AI-based web application that can provide decision-support recommendations based on farmer-provided information.

3. PROPOSED SOLUTION

The proposed system is an AI Farmer Decision Assistant that provides agricultural recommendations through a simple web interface. Farmers can enter information such as location, crop, soil type, water availability, weather condition, crop growth stage, and farming objective. The system analyzes these inputs and generates recommendations related to crop selection, irrigation, fertilizer guidance, crop care, and possible problems.

4. OBJECTIVES

- Develop an AI-assisted agricultural decision-support web application.
- Provide simple farming recommendations.
- Help farmers select suitable crops based on basic conditions.
- Provide irrigation and fertilizer guidance.
- Assist with basic crop-problem identification.
- Make agricultural information easier to understand.
- Demonstrate the use of AI-assisted development for a practical real-world problem.

5. SCOPE OF THE PROJECT

The current project focuses on crop recommendations, irrigation guidance, fertilizer guidance, basic crop-problem assistance, weather-related farming suggestions, and crop-care recommendations. Future versions can include real-time weather APIs, live market prices, soil sensors, crop disease image detection, voice interaction, Telugu and other regional languages, government scheme information, location-based recommendations, and personalized farmer profiles.

6. TECHNOLOGIES USED

The project is developed as an AI-assisted web application using the Base44 AI platform. The generated application may use modern web technologies such as HTML, CSS, JavaScript, and React-based components where applicable. Base44 AI is used to accelerate the creation of application pages, components, workflows, and functionality through

natural-language instructions.

7. SYSTEM REQUIREMENTS

Hardware requirements: computer or laptop, minimum 4 GB RAM, keyboard and mouse, and internet connection.
Software requirements: modern web browser such as Google Chrome, Microsoft Edge, or Firefox; Base44 AI platform; and internet connectivity.

8. SYSTEM ARCHITECTURE

The system follows a simple input-analysis-recommendation architecture. The farmer interacts with the web application, enters agricultural information, and submits the form. The AI-assisted recommendation logic analyzes the inputs and produces personalized decision-support suggestions. The recommendations are then displayed to the farmer in a structured format.

9. MAIN MODULES

Home Page: Introduces the project and provides navigation.
Farmer Input Module: Collects crop, soil, location, water availability, weather, and growth-stage information.
Crop Recommendation Module: Suggests potentially suitable crops based on provided conditions.
Irrigation Recommendation Module: Provides basic irrigation guidance.
Fertilizer Recommendation Module: Provides general nutrient-management suggestions.
Crop Problem Assistance: Provides possible crop problems, symptoms, and preventive guidance.
AI Recommendation Module: Combines farmer inputs and generates understandable recommendations.

10. WORKING OF THE SYSTEM

The farmer opens the application, enters the required agricultural information, and submits the form. The system processes the information through its AI-assisted recommendation logic. It then generates relevant recommendations and displays them in a simple format. The farmer can use the results as decision-support information and, where necessary, verify important decisions with local agricultural experts.

11. USER INTERFACE

The application is designed to be simple and farmer-friendly. Important interface features include clear navigation, readable text, easy-to-use forms, structured recommendation cards, clear buttons, responsive layouts, and minimal technical terminology. The interface can be extended with regional-language and voice support.

12. BASE44 AI IMPLEMENTATION

Base44 AI is used as the primary AI-assisted development platform. The application can be generated and refined using natural-language prompts. The tool can assist with creating pages, forms, layouts, workflows, and application logic.

13. ADVANTAGES

The system is easy to use, provides quick access to decision-support information, supports personalized suggestions, uses AI-assisted logic, provides a simple interface, can be accessed through a web browser, and can be extended with real-time APIs, regional languages, and voice interaction.

14. LIMITATIONS

Recommendations depend on the quality of farmer-provided information. AI-generated recommendations may not always be completely accurate. The prototype may not have direct access to real-time soil or weather data. Live market information requires external data sources. Accurate crop disease diagnosis may require image-based AI. Internet connectivity may be required. The system should be treated as a decision-support tool and not as a replacement for professional agricultural advice.

15. FUTURE ENHANCEMENTS

Future development can include real-time weather API integration, live agricultural market prices, voice-based interaction, Telugu and other regional-language support, crop disease detection using computer vision, IoT soil sensors, soil-moisture monitoring, personalized farmer profiles, and government agricultural scheme information.

16. TESTING

The application should be tested with different combinations of farmer inputs. Test cases should verify valid input submission, missing-field validation, different soil and crop selections, irrigation recommendations, responsive behavior on mobile screens, invalid inputs, and complete recommendation generation.

17. SAMPLE USE CASE

A farmer enters Location: Andhra Pradesh, Crop: Tomato, Soil Type: Loamy, Water Availability: Medium, Weather: Warm, and Growth Stage: Flowering. The assistant may respond with general advice such as maintaining appropriate irrigation, avoiding excessive watering, monitoring the crop for pests or disease, considering local weather before irrigation, and consulting local agricultural experts or soil-test results for fertilizer and disease-treatment decisions.

18. EXPECTED OUTCOME

The expected outcome is a functional AI-assisted web application that provides farmers with simple agricultural recommendations. The project demonstrates how AI-assisted development platforms can be used to create practical solutions for real-world agricultural problems.

19. CONCLUSION

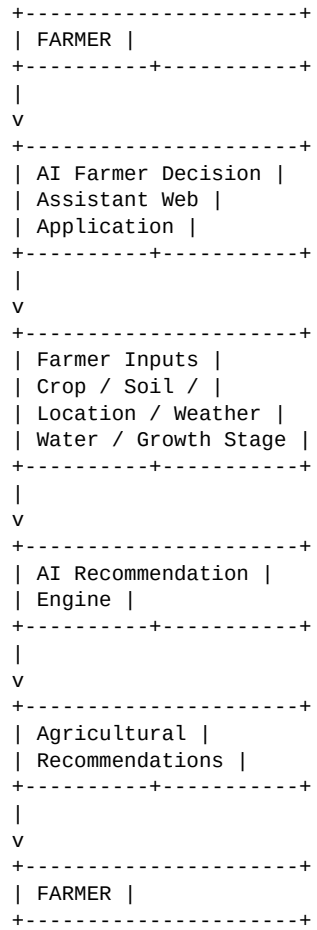
The AI Farmer Decision Assistant is a practical AI-assisted web application designed to support farmers in making better agricultural decisions. It collects basic information and provides understandable suggestions related to crop selection, irrigation, fertilizer management, crop care, and possible agricultural problems. By using Base44 AI, the

development process can be accelerated through natural-language application generation. The prototype provides a foundation for future improvements such as real-time weather, market prices, IoT sensors, computer vision, voice interaction, and regional-language support.

20. REFERENCES

1. Base44 AI – AI-assisted application development platform.
2. General agricultural information and farming practices.
3. Artificial Intelligence and Machine Learning concepts.
4. Web application development concepts.
5. Weather and agricultural data APIs for future integration.

SYSTEM ARCHITECTURE DIAGRAM



BASE44 AI SAMPLE PROMPT

```
Create a modern and simple web application called
"AI Farmer Decision Assistant".

The application should help farmers make better
agricultural decisions using AI.

Create a clean farmer-friendly homepage with:
- Project title
- Short description
- Get Started button
- Simple navigation

Create a farmer input form with:
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- Location
- Crop
- Soil type
- Water availability
- Weather condition
- Crop growth stage
- Farming objective

After submission, generate an AI-powered recommendation
page displaying crop recommendation, irrigation advice,
fertilizer guidance, crop-care suggestions, possible crop
problems, and preventive measures.

Use a clean, responsive, mobile-friendly design with
clear cards, icons, buttons, and readable text.
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TEST CASE TABLE

Test Case	Input / Condition	Expected Result
TC01	Valid crop and soil	Recommendation displayed
TC02	Missing crop	Validation message
TC03	Valid water availability	Irrigation advice displayed
TC04	Different soil type	Recommendation adapts
TC05	Different crop	Crop-specific advice
TC06	Mobile screen	Responsive interface
TC07	Invalid input	Error/validation message
TC08	Complete details	Full recommendation generated